

Risk and Opportunity Matrix - Sprint 2 (v2)

Introduction

In Sprint 2, the group revisited the risk and opportunity matrix produced in Sprint 1 and updated it with the evidence collected during the multi-source EDA, the Payoff Matrix construction, and the Data Characterization Report.

This version preserves the same structure used in Sprint 1 (risks and opportunities separated, identical column set, same grading scale) and only introduces three additions required by the Sprint 2 rubric: an Executive Summary section, an explicit Status column distinguishing Unchanged / Updated / New items, and a written justification for every item whose probability, impact, or mitigation has changed since Sprint 1.

The matrix continues to be a living document. Items closed out by Sprint 2 evidence remain in the register with the reduced grade rather than being removed, so the audit trail across sprints stays intact.

Executive Summary

At the conclusion of Sprint 2, the overall risk profile of the Bayer Decision Agriculture project has shifted from data structural uncertainty to analytical interpretation and data governance. During Sprint 1 the primary concern was the structural integrity of the ~320k planting rows; the multi-source EDA (Notebook 02) neutralized this threat by proving 100% join coverage between planting, operations, and harvest records and a 0.03% area-yield inconsistency rate. As a result, R1 and R3 dropped from Critical to Low.

A major analytical pivot also occurred: the original opportunity to find strong predictive correlations through the Agronomic Fitness Index failed validation (Cramér's $V = 0.0035$). The team pivoted the opportunity to a Yield-Grounded Scenario Framework (Pessimistic < P25, Baseline, Optimistic > P75), which now feeds the Sprint 2 Payoff Matrix.

The new dominant risks are therefore analytical and procedural: avoiding unjustified payoff values (R7), avoiding the bimodal corn/soy density mixture trap (R8), handling the missing productivity unit of measurement (R9), and the escalation of the retroactive GitLab management penalty (R5) into a categorical-zero scenario under the Sprint 2 rubric. The project remains highly viable; mitigation effort now pivots from data cleaning to rigorous agronomic interpretation, defensible payoff justification, and disciplined sprint-process hygiene.

Risks

Status	Risk	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Risk Response	Action Plan (Summary)	Owner / Function	Date Identified
Updated	Inconsistencies in the core dataset (R1)	10%	Low	Low	EDA 02 proved 100% join coverage across planting, operations and harvest, with only 0.03% area-yield inconsistency. The threat is largely neutralized but kept under monitoring.	Mitigate	Treat the dataset as the trusted base for the Payoff Matrix; no further imputation required. Re-check if Sprint 3 introduces new joins.	Arthur Almeida & João Cauê / Data Analysis	29/Apr/2026 (updated 14/May/2026)
Unchanged	API rate limits or downtime (R2)	40%	High	Elevated	Relying on external public sources (INMET/ Sentinel) introduces a technical dependency on third-party uptime for the simulator's inputs.	Mitigate	Retain the static historical sample as a local cache/ mock backend for Sprint 3 development.	Yan Kruziski / Risk Matrix	29/Apr/2026

Status	Risk	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Risk Response	Action Plan (Summary)	Owner / Function	Date Identified
Updated	Misalignment in Problem Framing (R3)	10%	Low	Low	Sprint 2 locked the decision variables to Planting Density and Spraying Windows, validated against the EDA. The risk shifts from open-ended alignment to drift prevention.	Mitigate	Hold the scope unchanged through the rest of Sprint 2; re-validate with Bayer only if a stakeholder request explicitly reopens the decision variables.	João Agmont / Problem Framing	29/Apr/2026 (updated 14/May/2026)
Unchanged	Superficial PESTEL Analysis (R4)	40%	Moderate	Relevant	Delivering generic macro factors without mathematically connecting them to the ROI of the planting strategy reduces the artifact's utility.	Avoid	Each PESTEL factor must include (1) quantified market sizing, (2) at least one specific regulatory citation, and (3) a financial impact estimate tied to Bayer's planting strategy ROI. Cross-review with Framing kept as a secondary check.	Raphael Skitnevsky / PESTEL	29/Apr/2026

Status	Risk	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Risk Response	Action Plan (Summary)	Owner / Function	Date Identified
Updated	Retroactive GitLab management penalties (R5)	80%	Critical	Critical	The Sprint 2 rubric introduced a categorical penalty: issues created and closed on the same day at the end of the sprint produce an automatic zero on the Individual Planning Factor (20% of the individual grade). The failure mode is no longer graded but binary.	Mitigate	Daily verification (mandatory) that every active task is created, moved to “Doing”, and closed on different days, with evidence captured before each daily check-in.	Miguel Almeida / GitLab Management	29/Apr/2026 (updated 14/May/2026)

Status	Risk	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Risk Response	Action Plan (Summary)	Owner / Function	Date Identified
Unchanged	Scope creep toward legacy systems (R6)	20%	High	Relevant	The team might waste resources attempting to integrate the prototype with Bayer's internal systems, which is expressly out of scope.	Avoid	Prevention: out-of-scope boundaries documented in the Problem Framing deliverable. Contingency: any incoming scope request is logged as a scope-evaluation issue, evaluated same-day by the Framing owner, and deferred to the next sprint planning if not accepted, preserving an audit trail.	João Agmont / Problem Framing	29/Apr/2026

Status	Risk	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Risk Response	Action Plan (Summary)	Owner / Function	Date Identified
New	Unjustified Payoff Values (R7)	40%	Critical	Critical	The Sprint 2 rubric penalizes payoff values invented without a documented source. The dataset has no localized financial data (R\$), so any fabricated profit margin would invalidate the Payoff Matrix.	Avoid	Anchor every payoff cell to the validated <code>average_yield</code> (kg/ha) metric from EDA 02. Each row of the matrix must cite the EDA section that produced its number.	Arthur Almeida & João Cauê / Data Analysis	14/May/2026
New	Cross-Crop Density Mixture Trap (R8)	30%	High	Elevated	Seed density in the Bayer database is strongly bimodal (corn ~70k seeds/ha vs soybeans ~300k seeds/ha). Averaging across crops produces a mathematically valid but agronomically nonsensical recommendation.	Avoid	Scope every Payoff Matrix alternative to a single crop (Soybeans-only for the v1 artifact). Corn deferred to a follow-up matrix.	Arthur Almeida & João Cauê / Data Analysis	14/May/2026

Status	Risk	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Risk Response	Action Plan (Summary)	Owner / Function	Date Identified
New	Missing productivity_uom (Data Governance) (R9)	50%	High	Elevated	100% of the 89,902 harvest rows have an empty productivity_uom field. The kg/ha assumption from the data dictionary is unverified and propagates into every payoff cell.	Mitigate	Tag every simulator input with a YIELD_UNIT_UNKNOWN flag during ingestion and document the kg/ha assumption explicitly in both the Data Characterization Report and the Payoff Matrix narrative.	Arthur Almeida & João Cauê / Data Analysis	14/May/2026

Opportunities

Status	Opportunity	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Opportunity Response	Action Plan (Summary)	Owner / Function	Date Identified
Updated	High correlation findings in EDA (O1)	100%	High	Elevated	The original Agronomic Fitness Index correlation failed (Cramér's $V = 0.0035$), but the same harvest dataset yielded a clean scenario partition through Yield Percentile Tiers (P25 / Baseline / P75) — a stronger, evidence-grounded substitute.	Exploit	Use the Yield Percentile Tiers as the official “States of the World” axis of the Sprint 2 Payoff Matrix and as the scenario input for the Sprint 3 decision logic.	João Cauê / Data Analysis	29/Apr/2026 (updated 14/May/2026)
Unchanged	Strategic differentiation via PESTEL (O2)	60%	High	Elevated	A deep, quantified analysis of Brazilian agribusiness volatility provides a unique angle for the Payoff Matrix narrative.	Enhance	Use economic volatility to justify Savage/Minimax-Regret choices in the Payoff Matrix narrative; quantify market sizes and regulatory impacts in the PESTEL document.	Ariel Mouadeb / PESTEL	29/Apr/2026

Status	Opportunity	Prob. (%)	Impact	Grade / Intensity	Fundamental Justification	Opportunity Response	Action Plan (Summary)	Owner / Function	Date Identified
Unchanged	Establishing a reusable Risk Framework (O3)	80%	Moderate	Relevant	Building a highly structured matrix in Sprint 1 accelerated the mandatory update for Sprint 2 — this v2 artifact is the first realization of that opportunity.	Exploit	Keep the matrix mapping directly tied to the workflow, allowing rapid status updates for Sprint 3.	Yan Kruziski / Risk Matrix	29/Apr/2026

Justifications for Updated and New Items

R1 — Inconsistencies in the core dataset · Reduced from Critical to Low

In Sprint 1 this risk was rated 60% × Very High = Critical. EDA 02 (Notebook 02) revealed that 100% of the planting events map directly to a harvest and operations record, with zero orphaned records, and the Area-Yield consistency check returned only a 0.03% inconsistency rate. The probability of core data inconsistency breaking the simulator is therefore drastically reduced. The risk is retained in the register for monitoring purposes only, not because new mitigation is required.

R3 — Misalignment in Problem Framing · Reduced from Critical to Low

Originally 30% × Very High = Critical. Sprint 2 locked the scope on two specific decision variables — Planting Density and Spraying Windows — validated against the EDA and accepted by the Framing owner. With the decision surface explicit and bounded, the residual risk shifts from open-ended alignment to drift prevention, and the grade drops accordingly.

R5 — Retroactive GitLab management penalties · Escalated to Critical

Originally 50% × High = Elevated. The Sprint 2 rubric introduced a sharper rule: the Individual Performance Factor allocates exactly 20% to Planning, and the rule now reads “if issues are created and closed on the same day at the end of the sprint, your planning score is automatically 0.” This converts what was a graded deduction into a categorical zero, raising the impact to Critical. Probability is raised to 80% based on observed team behavior in the final days of Sprint 1.

O1 — High correlation findings in EDA · Pivoted

The exploitation strategy planned in Sprint 1 — using the Agronomic Fitness Index to seed the decision tree — failed validation (Cramér's $V = 0.0035$, statistically indistinguishable from noise). Instead of abandoning the opportunity, the team pivoted: the same harvest dataset supports a defensible scenario partition through Yield Percentile Tiers (Pessimistic < P25, Baseline P25-P75, Optimistic > P75). These tiers now serve as the official “States of the World” axis of the Sprint 2 Payoff Matrix, a stronger evidence-grounded substitute for the original correlation play.

R7 — Unjustified Payoff Values · New

The Sprint 2 rubric is explicit: payoff values invented without a documented source are penalized, and a single un-sourced cell can invalidate the artifact. The dataset contains no localized financial data (R\$), so any attempt to fabricate profit margins per hectare would fail the requirement. Mitigation: anchor every payoff cell to the validated `average_yield` (kg/ha) metric derived from the harvest dataset, and cite the EDA section that produced each number directly in the Payoff Matrix.

R8 — Cross-Crop Density Mixture Trap · New

EDA 01 and 02 proved that seed density in the Bayer database is strongly bimodal due to the mixture of corn (~70k seeds/ha) and soybeans (~300k seeds/ha). If the team builds a Decision Matrix alternative on “average seed density” without isolating the crop, the resulting Laplace or Maximax recommendation will be mathematically valid but agronomically nonsensical, recommending a density that exists for neither crop in practice. Mitigation: every Payoff Matrix scenario in Sprint 2 must declare its crop scope (Soybeans-only for the v1 artifact), with corn deferred to a follow-up matrix.

R9 — Missing productivity_uom (Data Governance) · New

EDA 02 uncovered a silent data-quality trap: 100% of the 89,902 harvest rows have an empty `productivity_uom` field. The data dictionary states the values are in kg/ha, but this is an unverified assumption that propagates into every payoff cell. If Bayer later confirms a mixture of bags/ha or bushels/acre in the source feed, the simulator's output would be wrong by an order of magnitude. Mitigation: tag every simulator input with a `YIELD_UNIT_UNKNOWN` flag during ingestion and document the kg/ha assumption explicitly in both the Data Characterization Report and the Payoff Matrix narrative, so any contradicting evidence later can be reconciled without rework.

Top 3 Risks - Prioritization

Severity grade alone does not capture how a risk will behave in the next sprint. The three risks below are prioritized using two additional criteria beyond probability × impact: (a) propagation cost — how many downstream deliverables inherit the failure, and (b) recovery asymmetry — whether the failure is gradual and partially recoverable, or categorical and unrecoverable without rework.

1. R7 — Unjustified Payoff Values

R7 directly threatens the central Sprint 2 deliverable. The Payoff Matrix is the artifact the rubric grades most strictly, and every payoff cell is a separate point of failure. The failure mode is categorical: a single fabricated number invalidates the rationale section. Recovery requires returning to the EDA, re-deriving every cell, and rewriting the narrative — a cost that does not fit in the remaining Sprint 2 budget.

2. R5 — Retroactive GitLab management penalties

The escalation to Critical and the categorical-zero rule make this the highest-probability disaster path. Unlike technical risks, R5 cannot be fixed by a single deliverable: it requires daily discipline across the whole team. The deficit compounds across the module because the Individual Performance Factor is evaluated cumulatively and cannot be backfilled once the sprint window closes.

3. R8 — Cross-Crop Density Mixture Trap

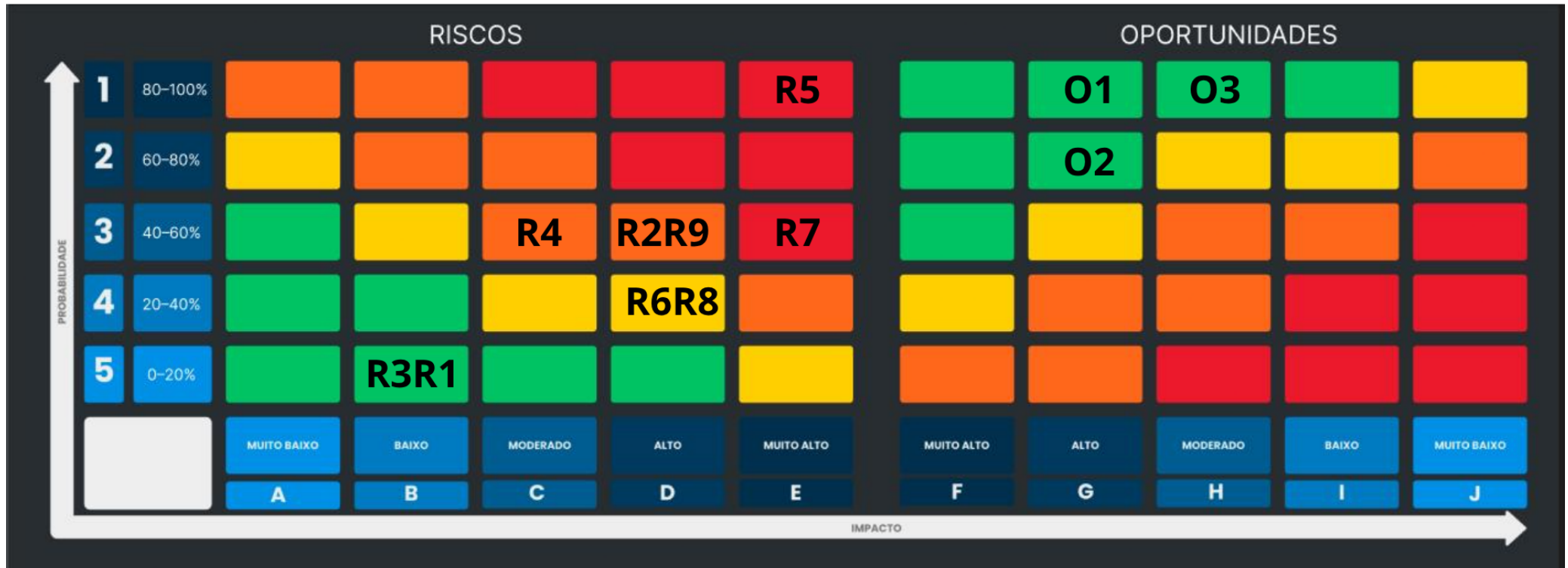
R8 sits below R7 on grade but ranks above the remaining risks because its failure mode is silent: a bimodality-induced wrong recommendation looks numerically clean and will pass any unit test. The Payoff Matrix would be internally consistent but agronomically invalid — a failure that survives review and only surfaces when an actual agronomist reads the recommendation. There is no cheap downstream control that catches it.

Why not R9

R9 is High impact but its mitigation is bounded and cheap (one ingestion flag, one documented assumption). Once the flag and disclaimer are in place, the risk is contained even if the assumption later proves wrong. R7, R5, and R8 all require sustained attention through the end of Sprint 2; R9 does not.

Visual Map

The Sprint 2 heat map below uses the same Probability × Impact axes adopted in Sprint 1 (Probability rows 1-5, Impact columns A-J, with the opportunity side mirrored so the higher-impact opportunities sit closer to the centre).



Sprint 2 risk and opportunity heat map

Cell placement summary:

Cell	Items	Reading
1E	R5	Critical · 80% probability · the dominant process risk
3E	R7	Critical · 40% probability · the dominant analytical risk
3D	R2, R9	Elevated · 40–50% probability · technical and governance risks
3C	R4	Relevant · 40% probability · PESTEL depth risk
4D	R6, R8	Elevated · 20–30% probability · scope and bimodality risks
5B	R1, R3	Low · 10% probability · resolved in Sprint 2, kept for monitoring
1G	O1	Elevated · 100% · pivoted exploitation via yield percentile tiers
1H	O3	Relevant · 80% · reusable risk framework
2G	O2	Elevated · 60% · strategic differentiation via PESTEL

Sprint 2 movements visible on the map:

- R1 and R3 collapsed from the top-right quadrant (Critical in Sprint 1) to 5B (Low) after EDA 02 evidence neutralized both.
- R5 moved up to 1E (Critical) under the new categorical-zero rule.
- O1 stays in the Elevated band but its underlying mechanism changed (correlation replaced by yield percentiles).
- Three new items populate previously empty cells: R7 at 3E, R9 at 3D, and R8 at 4D.

Mitigation Action Confirmation

ID	Mitigation status	Owner	Next review
R1	Action complete	Arthur Almeida & João Cauê	Monitoring during Sprint 3
R2	Action active	Yan Kruziski	Sprint 3 kick-off
R3	Action complete	João Agmont	Sprint 3 kick-off
R4	Action active	Raphael Skitnevsky	Mid-Sprint 3
R5	Action active (daily)	Miguel Almeida	Daily
R6	Action active	João Agmont	On scope-change trigger
R7	Action active	Arthur Almeida & João Cauê	Before Sprint 2 submission
R8	Action active	Arthur Almeida & João Cauê	Before Sprint 2 submission

ID	Mitigation status	Owner	Next review
R9	Action active	Arthur Almeida & João Cauê	Before Sprint 2 submission and Sprint 3 kick-off
O1	Exploitation active	João Cauê	Sprint 3 kick-off
O2	Exploitation active	Ariel Mouadeb	Mid-Sprint 3
O3	Exploitation active	Yan Kruziski	Each sprint start

Conclusion

Sprint 2 resolved the two largest Sprint 1 risks (R1, R3) through EDA evidence rather than through optimism. In exchange, the project inherits a different class of risk: interpretation discipline in the Payoff Matrix (R7, R8), data governance around unit conventions (R9), and a sharper version of the recurring process risk (R5).

The matrix will be revisited at Sprint 3 kick-off, when the simulator implementation phase begins and a new wave of risks — model overfit, UI usability, and stakeholder validation — is expected to surface. As in Sprint 1, items closed by evidence will remain in the register at their reduced grade rather than being removed, preserving the audit trail required by Inteli's evaluation.